AI-Augmented Engineering and Operations for Avion's Full-Flight Simulator Fleet

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***Abstract*—This paper proposes an AI platform, grounded in Avion's own engineering documents, that helps scale full-flight simulator engineering and operations without proportional headcount growth. In Phase 1 propose/approve, discipline-specific AI agents search authorised document libraries, cite their sources, and draft outputs for human approval. Eight near-term initiatives (P1 – P8) prioritise documentation, diagnostics, audit readiness, release integrity, instructor workflows, and operational knowledge transfer. A 90-day pilot at London Luton can validate the approach.**

I. Company context and competitive signals

Avion already differentiates through high reliability (99.7% uptime, over 5,000 annual training hours at London Luton [2]) and lower maintenance burden than traditional simulators [1]. As Level D capability becomes more widely available, including from newer entrants such as Indra [9] and HAVELSAN [10], differentiation increasingly shifts to **operational intelligence**. Table I summarises public competitor AI/data signals motivating why now; P1–P8 position Avion to respond before competitors define the category. Appendix A provides the annotated public source pack, including contextual hardware-parity signals.

TABLE I
Competitor AI/data product signals.

Vendor	Key initiative / product
CAE	Automated event detection + real-time insights (Rise) [5]
FlightSafety	AI/ML performance evaluation (FlightSmart) [6]
Simaero	Competency-Based Training and Assessment (CBTA) platform + real-time data feedback (Hinfact) [7]
Thales	PERCEVAL competency evaluation programme [8]

II. AI operational model

The goal is not a general-purpose AI. I propose a single platform, grounded in Avion's own documents, that hosts multiple **discipline-specific AI agents**. The approach is intentionally incremental, aligned with FAA [12] and EASA [13] guidance, so value is delivered quickly while autonomy expands only after measured performance. This pattern is visible in adjacent industries: Siemens' Industrial Copilot [17] uses AI agents that ingest automation and process-simulation data to assist engineers with code generation and maintenance instructions, and Airbus' Skywise platform [18] connects over 12,000 aircraft for AI-driven predictive maintenance and operational analytics.

A. Discipline-specific AI agents

The platform is organised around **AI agents**, each dedicated to one engineering or operational discipline. Every agent:

- has access **only** to the documents and tools relevant to its own discipline,
- can query other disciplines' libraries when a task spans boundaries,
- logs cross-discipline retrievals and includes them in the approval bundle,
- cites the source document for every claim it makes.

In the intended operating mode, agents can run continuously: they monitor their document libraries for changes, flag inconsistencies or gaps as they appear, and keep draft outputs up to date so that engineers always have a current starting point rather than a blank page. Agents can also proactively surface relevant information within their scope, subject to the approval rules defined in Phase 1. Continuous operation means maintaining drafts and raising flags, not modifying source systems or released artefacts.

Table II lists the initial set of agents and the document types each one draws from.

TABLE II
Discipline-specific AI agents and their primary document scope.

Agent	Primary document scope
Systems engineering	Requirements baselines, interface control documents, change requests, trace links
Simulation software	Simulator runtime, flight-model integration, visual pipelines, flight-performance data
Test & qualification	FAA Part 60, EASA CS-FSTD, Qualification Test Guide (QTG) procedures, qualification packages
Site operations	Daily readiness, incidents, maintenance notes, instructor SOPs, centre commissioning
Training / instruction	EBT/CBTA scenario libraries, instructor briefing material, competency rubrics, and recurrent training records
Field support	Ticket triage, known-issue knowledge base, root-cause summaries, draft patches
Electrical / avionics	Wiring diagrams, avionics integration specs, component datasheets
Mechanical / motion	Motion system maintenance, hydraulic/electric actuator logs, alignment procedures
Quality management	ISO/AS audit checklists, non-conformance reports, Corrective and Preventive Action (CAPA) records
DevOps	Build configurations, CI/CD pipelines, environment definitions, deployment logs
Product / go-to-market	Customer requirements, proposal templates, competitive positioning

TABLE III
CURRENT PAIN POINTS AND PROPOSED AI SOLUTIONS.

Pain point	Initiative	Impact	Effort
Slow, inconsistent documentation	P1: Technical documentation	High	Med
Reactive monitoring	P2: Simulator fleet diagnostics and triage	High	Lg
Fragile traceability	P3: Change-impact agent	High	Med
Evidence assembly bottleneck	P4: Evidence packager	High	Med
Reactive CI/CD, no flight-perf pipelines	P5: Secure DevOps copilot	Med–Hi	Med
Unstandardised instructor workflows	P6: Instructor workflow agent	Med–Hi	Med
Dependency on tribal knowledge & slow onboarding	P7: Operational playbooks & onboarding	Med–Hi	Sm
Release provenance gaps	P8: Release integrity	Med–Hi	Med

Effort legend: Sm = Small, Med = Medium, Lg = Large.

B. Phased autonomy and governance

- **Phase 1 (propose/approve):** agents draft cited outputs; **humans approve** every material action.
- **Phase 2 (routine execution):** agents execute repeatable, low-risk workflows within explicit tool constraints; humans review exceptions and spot-check samples.
- **Phase 3 (workflow ownership):** agents own end-to-end repeatable workflows with auditable logs and measurable error rates; humans focus on oversight and regulatory sign-off.

III. PRIORITY INITIATIVES (P1–P8)

To prove value in Phase 1, the near-term roadmap focuses on eight initiatives (P1 – P8) mapped to recurring pain points (Table III). Each initiative also inherits risks intrinsic to AI agents—hallucination, misattribution, cross-domain interpretation errors, and prompt injection. These are not listed per initiative because they apply uniformly; instead, Appendix C defines the reliability control matrix that every agent must satisfy before its outputs reach review.

A. Theme 1: Faster engineering

P1: Technical documentation: **Challenge.** Technical documents (system descriptions, test procedures, manuals, qualification reports) are maintained manually across OEM and internal sources. This causes duplication, inconsistent wording, slow updates, and stale artefacts reaching review.

Proposed solution: Technical documentation agent. An agent drafts and updates documents using Avion templates and style guides with per-paragraph citations. When upstream OEM content changes, it flags affected documents and proposes updates; engineers review and approve.

P2: Simulator fleet diagnostics and triage: **Challenge.** Without end-to-end correlation and consistent runbooks, monitoring becomes reactive: faults trigger manual correlation across subsystem logs and separate OEM troubleshooting guides, so diagnosis depends on individual experience [1].

Proposed solution: Diagnostics and triage agent. Standardise logs/metrics/traces collection and correlation using industry-standard observability tooling so an agent can detect anomalies, retrieve relevant fault-isolation procedures, and draft an incident report with ranked hypotheses.

P5: Secure DevOps copilot: **Challenge.** Build failures, environment drift, and security vulnerabilities consume engineering time. Flight-model and performance-data workflows also lack repeatable build-and-test pipelines.

Proposed solution: Secure DevOps copilot. Automate build-failure diagnosis, reduce environment drift, and enforce a consistent security baseline (dependency updates, secrets detection, least-privilege access). Extend automated pipelines to flight-model and performance-data workflows so that tuning and validation are repeatable rather than manual.

B. Theme 2: Audit-ready operations

P3: Traceability and change-impact analysis: **Challenge.** Requirements, baselines, and evidence are fragmented across tools; change-impact analysis is manual and incomplete. This increases audit risk under FAA Part 60 and EASA CS-FSTD [3], [4].

Proposed solution: Change-impact agent. Maintain bidirectional trace links and automatically draft impact assessments (affected tests, evidence, baselines, and checklists) for engineering review, strengthening configuration control and audit trails [3], [4].

P4: Qualification evidence assembly: **Challenge.** Each release requires an evidence packet (test results, logs, configuration snapshots, approvals, deltas). Manual collation causes omissions and qualification delays [3], [4].

Proposed solution: Qualification evidence packager. Automatically collate artefacts into an audit-ready evidence packet with tamper-evident verification, provenance pointers, and a summary. The packager does not transform objective-test data; it collates, hashes, and indexes evidence while preserving QTG presentation requirements [3], [4].

P8: Release integrity: **Challenge.** Without verifiable proof of what went into each release, supply-chain risk rises, audits slow, and root-cause investigations become harder.

Proposed solution: Release integrity pipeline. Automatically produce signed release packages with a complete inventory of included components and a tamper-evident record of how each artefact was built, following approved release-provenance and signing controls.

C. Theme 3: Training and knowledge transfer

P6: Instructor workflows and scenario authoring: **Challenge.** Session preparation, scenario authoring, and debriefs depend on individual instructor practice. Objective event definitions and competency mapping are not consistently standardised, limiting grading consistency and reuse. Industry direction toward Evidence-Based Training (EBT) and Competency-Based Training and Assessment (CBTA) [11] is accelerating, as reflected in the competitor signals in Table I.

Proposed solution: Instructor workflow agent. Standardise session setup, generate scenario scripts from syllabus parameters, and draft debriefs based on **structured, objective events** linked to EBT/CBTA competency markers. Output is structured scenario and debrief artefacts rather than free-form narrative.

P7: Operational playbooks and onboarding: **Challenge.** Commissioning new simulation centres and onboarding new staff depend on scattered procedures and senior-engineer tribal knowledge.

Proposed solution: Operational playbooks agent. An agent generates site checklists, surfaces the right OEM/regulatory procedure at each stage, tracks completion, and provides step-by-step guidance. This enables Avion to scale operations and open new centres without linearly scaling the specialist workforce.

IV. Medium-term roadmap (6–18 months)

These capabilities extend the P1–P8 foundation as autonomy increases:

- **Digital thread:** assign every artefact a unique ID and link it through the full lifecycle (documentation, configuration, build, test, evidence, operations). This makes it possible to trace any change back to its origin and, if needed, revert to a previous qualified baseline, as required by CS-FSTD configuration management guidance [4].
- **Objective training analytics:** move beyond scenario generation (P6) to automated grading, longitudinal trend analysis, and standardised competency artefacts.
- **Predictive maintenance:** use fleet-wide operational data to build reliability models that predict component wear and schedule maintenance proactively, reducing unplanned downtime.

V. Deployment and security

A. Data sovereignty and deployment tiers

The central design principle is **data sovereignty**. Workloads are tiered by sensitivity; Appendix C details the full control set.

Tier 1 (Offline): OEM documents, qualification evidence, and operational logs run on **Avion-maintained on-prem infrastructure** with no internet path.

Tier 2 (Controlled): Low-sensitivity tasks such as generic drafting may use controlled outbound connectivity to cloud LLM APIs under contracted enterprise controls that prohibit model training on Avion data and enforce zero data retention where eligible.

Across both tiers, **human oversight is mandatory**, consistent with EASA and FAA AI guidance [13], [12]. Outputs remain source-grounded, and each agent can access only the tools it needs.

B. Governance

We threat-model LLM-specific risks and govern them using industry frameworks mapped in Appendix C: NIST AI RMF [15], NIST SP 800-53 [16], and OWASP LLM Top 10 [14].

C. Infrastructure and cost

The platform runs on Avion-managed compute and storage with a site LAN connection to simulator hosts. A proof of concept can reuse existing infrastructure where available; final sizing is determined during pilot scoping (Appendix B). For planning only, Appendix D provides an order-of-magnitude infrastructure and API cost envelope using public list pricing.

VI. Recommendation

The strategic objective is simple: **decouple simulator fleet growth from headcount** using the Phase 1 operating model where agents propose and humans approve. To validate this, I propose a **90-day pilot** at London Luton [2], shipping P1, P3, and P4 on one aircraft program, with P2 included where telemetry readiness permits. Pilot scoping options are in Appendix B.

Success will be measured against baseline values for three indicative targets:

- **50% faster diagnosis** on telemetry-enabled scope,
- **50% lower audit-preparation effort**,
- **>20 engineering hours saved per week** across the pilot team.

These targets will be refined during pilot planning and used to inform a company-wide rollout decision. Resourcing and infrastructure are intentionally scoped during that planning to avoid premature precision.

Appendix A
Annotated competitor source pack (public signals)

Purpose. This appendix provides source-level annotations for the competitor entries in Table I plus contextual hardware-parity signals. All items are public marketing and press materials, treated as **market signals**, not proof of deployed maturity.

A. Annotated source notes

CAE (Rise). Source: CAE civil aviation product material. https://www.cae.com/civil-aviation/aviation-software/cae-rise/cae-rise-data-analytics

- **Explicit claim (paraphrased):** automated event detection/assessment and real-time metrics from simulator data.
- **Operational implication for Avion:** mature event taxonomy and instructor analytics patterns relevant to P6 and, indirectly, P2.

FlightSafety (FlightSmart). Source: FlightSafety press release. https://news.flightsafety.com/pressrelease/flightsafety-introduces-flightsmart-new-integrated-pilot-performance-evaluation-training-tool-developed-conjunction-ibm/

- **Explicit claim (paraphrased):** integrated AI/ML evaluation + training workflow with objective measurement.
- **Operational implication for Avion:** validates market movement toward objective competency analytics (P6).

Simaero / Hinfact. Source: Simaero public announcement. https://www.sim.aero/news/simaero-partners-with-hinfact-to-power-its-transition-to-full-cbta

- **Explicit claim (paraphrased):** transition to full Competency-Based Training and Assessment (CBTA) with digital tooling and data feedback.
- **Operational implication for Avion:** indicates center-level CBTA workflow standardization pressure (P6/P7).

Thales (PERCEVAL). Source: Thales consortium announcement.
https://www.thalesgroup.com/en/worldwide/aerospace/press_release/perceval-thales-and-partners-reshape-future-airline-pilot

- **Explicit claim (paraphrased):** tooled competency and behavior evaluation aligned with DGAC/EASA direction.
- **Operational implication for Avion:** reinforces measurable competency artifacts and regulatory trajectory behind P6.

Indra (A320 Level D). Source: Indra announcement.
https://www.indracompany.com/en/noticia/indra-implements-second-airbus-a320-level-simulator-china

- **Explicit claim (paraphrased):** Level D simulator deployment activity.
- **Operational implication for Avion:** supports the "why now" argument that hardware parity pressure is increasing.

HAVELSAN (A320 Level D). Source: HAVELSAN announcement.
https://www.havelsan.com/en/news/airbus-a320-ffs

- **Explicit claim (paraphrased):** EASA Level D certification on A320 FFS deployment.
- **Operational implication for Avion:** confirms additional certified supply and supports differentiation via operational execution quality.

APPENDIX B
PILOT SCOPING OPTIONS

Purpose. This appendix provides scoped pilot options so management can choose validation depth before committing to wider deployment.

Scoping principle. Avoid false precision at this stage. Final staffing and hardware numbers depend on selected data sources, integration depth, and review cadence.

A. Pilot options at a glance

B. Capacity envelope

Rollout view (after pilot). If pilot criteria are met, a phased site-by-site rollout typically needs a small core team (2–3 engineers) plus local SMEs and IT support at each site.

C. Pre-launch checklist (by owner)

- **Engineering:** data scope (document formats, corpus volume, telemetry/log availability); workflow definition; acceptance criteria and technical ownership.
- **IT/Security:** hosting and network controls; identity/access controls and logging; tiering decision (offline-only vs. controlled outbound for low-sensitivity tasks).
- **Quality/Compliance:** evidence standards; audit acceptance criteria and sign-off boundaries.
- **Site operations:** integration scope (file shares, Git, CI, ticketing); SME validation bandwidth and runbook alignment.
- **Leadership/program office:** approval model (named approvers, correction latency, acceptance rules); dependency map against other digital/AI initiatives.

D. Pilot charter (evaluation template)

Primary metrics (baseline first): *Baseline definition:* use the most recent 8–12 comparable incidents/releases on the same program/device before pilot start.

- **Operational value:** diagnosis cycle time (median and 95th percentile, where telemetry is available), time-to-draft, and engineering hours saved.
- **Audit readiness:** effort/time to assemble release evidence packages (collation and traceability).
- **Reliability:** citation correctness, factual consistency vs. sources, approval/correction rate, and defect escape rate.

Suggested stop conditions:

- access or ingestion cannot be achieved for agreed scope,
- reliability thresholds remain below target after remediation cycles,
- no measurable progress by midpoint checkpoint.

Illustrative milestones:

- Weeks 1–2: access approvals, ingestion, and workflow definition.
- Weeks 3–6: P1/P4 on constrained scope with baseline capture.
- Weeks 7–9: extend with P3 and tighten traceability quality.
- Weeks 10–12: add P2 only if telemetry is ready; otherwise close instrumentation gaps, report P1/P3/P4 outcomes, and carry the diagnosis KPI to the next phase.

APPENDIX C
RELIABILITY AND SECURITY CONTROLS

Purpose. This appendix defines Phase 1 reliability and cybersecurity controls in auditable terms.

A. Reliability control matrix (Phase 1)

TABLE VI
RELIABILITY CONTROL MATRIX FOR PHASE 1 PROPOSE/APPROVE OPERATION.

Risk	Control mechanism
Hallucination / fabrication	**No citation, no claim.** Every material statement must include one or more citations to approved sources. If supporting source cannot be retrieved, the claim is rejected.
Incorrect citation / attribution	**Citation validation workflow.** Reviewers inspect cited passages during approval; automated checks verify source-to-claim mapping where possible.
Cross-domain interpretation	**Scoped retrieval by discipline.** Agents default to discipline-specific corpora; cross-domain access is explicit with preserved provenance.
Output inconsistency / weak structure	**Template-constrained outputs.** Use engineering templates (e.g., SB, TN, QTG evidence format) rather than unconstrained free text.
Prompt injection / unsafe tool usage	**Untrusted-content handling + tool gating.** Retrieved text is treated as untrusted. Tool calls are allowlisted, parameter-constrained, and human-approved for material actions.

TABLE IV
PILOT OPTIONS BY SCOPE, OBJECTIVE, DURATION, AND PLATFORM ENVELOPE.

Option	Objective	Scope	Indicative duration	Platform envelope
PoC (Option 1)	Validate retrieval, approvals, and citation-grounded drafting.	One agent (P1 Technical Documentation) on a limited approved corpus; draft-only workflow with engineering review.	2–4 weeks	See Appendix D for infrastructure sizing.
90-day Pilot (Option 2)	Validate operational and audit value on a live program.	P1, P3, and P4 on one aircraft program/device, with P2 where telemetry/logs are ready.	12 weeks (90 days)	See Appendix D for infrastructure sizing.

TABLE V
DIRECTIONAL CAPACITY SCENARIOS FOR POC/PILOT PLANNING (NOT COMMITMENTS).

Scenario	Best fit	Lead engineer	Domain SME review	Engineering support	IT / security / quality
Lean	Option 1 PoC (single workflow)	0.2–0.4 FTE	0.1–0.2 FTE	0.0–0.2 FTE	0.05–0.1 FTE
Standard	Option 2 pilot (P1/P3/P4 + P2 where ready)	0.5–0.7 FTE	0.2–0.4 FTE (combined)	0.3–0.5 FTE	0.1–0.2 FTE
Accelerated	Pilot + rollout-prep artifacts	0.7–1.0 FTE	0.3–0.6 FTE (combined)	0.5–0.8 FTE	0.2–0.3 FTE

B. Cybersecurity deployment tiers

TABLE VII
CYBERSECURITY DEPLOYMENT TIERS AND MANDATORY CONTROLS.

Tier	Scope and sensitivity	Mandatory controls
Tier 1 (Offline only)	Critical workloads: OEM documents, qualification evidence, release/sign-off artefacts, sensitive operational logs.	No internet route, Avion-managed on-prem environment, strong IAM and least privilege, signed artefacts, tamper-evident logging, full audit trail.
Tier 2 (Controlled online)	Selected low-sensitivity tasks: generic drafting and low-risk research support.	Allowlisted egress, data classification and redaction policy, enterprise no-training and zero-retention terms where eligible, least privilege, complete API and audit logging.

C. Control frameworks

- **NIST AI RMF 1.0** for AI-specific risk governance,
- **NIST SP 800-53** for identity, access, audit, and system integrity controls,
- **OWASP LLM Top 10** for prompt injection and insecure output handling mitigations.

APPENDIX D
COST AND INFRASTRUCTURE ENVELOPE

Purpose. This appendix fixes the planning envelope for the proposed delivery model: one Avion-owned on-prem inference box for Gemma 4 and one controlled external API path for Claude Opus 4.7. It is a budgetary guide, not a procurement quote.

A. Chosen pilot architecture

TABLE VIII
PILOT ARCHITECTURE AND PLANNING ENVELOPE.

Layer	Planning envelope
Tier 1 on-prem	Gemma 4 31B Dense served via vLLM on one workstation-class host with 24 vCPU, 128 GB ECC RAM, 4 TB NVMe, 10 GbE, and one NVIDIA RTX 6000 Ada (48 GB VRAM). Practical pilot capacity is one heavy analyst or two light concurrent users.
Tier 2 controlled API	Claude Opus 4.7 used only for low-sensitivity commercial drafting and synthesis. API calls are gated by policy review and used only after sanitisation.
Core platform	Python service layer, Qdrant for retrieval, PostgreSQL for workflow and audit metadata, and Langfuse/Prometheus/Grafana for traces and monitoring.

B. Budget envelope

TABLE IX
PILOT BUDGET ENVELOPE (PLANNING LEVEL).

Cost item	Envelope	Notes
On-prem hardware	EUR 15k–19k one-time	Workstation-class host, RTX 6000 Ada, ECC RAM, NVMe, and warranty/backup media allowance.
Core software	EUR 0 licenses	vLLM, Qdrant, PostgreSQL, Langfuse, Prometheus, and Grafana are open-source.
Claude Opus 4.7 API	USD 1.3k–2.6k for 90 days	Assumes roughly 200 requests per working day, with 10k–20k input tokens and 2k–4k output tokens per request at published list pricing [20].
Operations	EUR 600–900 per quarter	Power, monitoring, backup handling, and routine admin overhead.
Year-1 TCO	EUR 20k–30k	CapEx-heavy first year, with later years dominated by operations and API usage.

C. API pricing intuition

At the April 2026 public list price for Claude Opus 4.7 [20], the planning assumptions above give two simple anchor cases:

- **S1 (reviewed answer):** 10k input + 2k output tokens ⇒ about USD 100 per 1,000 requests.
- **S2 (document draft):** 20k input + 4k output tokens ⇒ about USD 200 per 1,000 requests.

That maps to a practical pilot envelope of roughly USD 400–900 per month at about 4.4k requests per month, depending on how draft-heavy the commercial workload becomes.

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